# NANYANG TECHNOLOGICAL UNIVERSITY

## SEMESTER 2 EXAMINATION 2024-2025

## SC2006/CZ2006 – SOFTWARE ENGINEERING

Apr/May 2025 Time Allowed: 2 hours

## INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer **ALL** questions.
3. This is an open-book examination.
4. All questions carry equal marks.
5. Refer to Appendix A on page 5 for the project description which is needed to answer some of the questions.

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1. Based on the project description given in Appendix A,

   (a) Identify FOUR key stakeholders for the Disaster Warning System (DWS) project and describe their primary interests and concerns related to the system.

   (4 marks)

   (b) Based on the project description, list THREE functional requirements and THREE non-functional requirements for the DWS.

   (6 marks)

   (c) Identify the actors and use cases of the project and draw the use case diagram. Use <<include>> and <<extend>> relationships where appropriate.

   (15 marks)

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2. (a) Draw a state machine diagram for the Sensor object in the DWS that shows the states and transitions of a Sensor object.

(6 marks)

(b) Draw an activity diagram with swim lanes if any, that shows the activities for the process of linking a sensor to an alert system.

(7 marks)

(c) A software development company is planning to build the DWS platform. The client has provided a detailed list of requirements but has also indicated that they may need to make changes to the requirements during the development process. The company is considering two process models for this project: the Waterfall model and the Agile model.

i. Based on the client's needs, which process model would you recommend, and why?

(3 marks)

ii. With your recommended process model how you will manage frequent changes in requirements?

(3 marks)

(d) You are the Scrum Master of the software development team working on the DWS platform. The team has just completed Sprint Planning and started the Sprint. During the Sprint, the Product Owner approaches you with a new high-priority feature request that they believe must be included in the current Sprint. Explain how you, as the Scrum Master, should handle this situation according to Scrum principles.

(6 marks)

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3. (a) Propose an appropriate architecture for the DWS as described in Appendix A, and briefly justify your choice.

(3 marks)

(b) The DWS needs to be able to register and de-register a wide variety of sensors during its runtime. Moreover, registered sensors can send measurements to the DWS at any time.

i. Name the design pattern that is most applicable to this problem statement and explain why it is appropriate.

(3 marks)

ii. Draw a detailed Class diagram with key attributes and methods for the DWS, reflecting the use of the design pattern from 3(b)i.

(10 marks)

(c) Your company developed and sold the DWS software, and has now been contracted to maintain it. Your client complains to you that they purchased a sensor from some other company and connected it to their DWS, but when the sensor unexpectedly sent an update to the DWS in the form of a string rather than the specified integer, this caused the DWS software to crash in an expensive and potentially dangerous outage.

i. Should your company be blamed for this outage? Give ONE reason why they should, and ONE reason why they should not.

(2 marks)

ii. Name TWO parties other than your company who could be considered to blame for this outage, briefly explaining why.

(4 marks)

iii. Describe THREE ways (at any stage of the development / maintenance process) in which your company could have either prevented this outage from happening, or reduced their risk of blame for the outage.

(3 marks)

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4. Attempting to link a sensor with an alert requires the following information:
   1. The name of the sensor in ASCII format.
   2. The name of the alert in ASCII format.
   3. A 5 digit integer N in decimal format with sign, representing the lower bound of the safe measurement range (e.g. +83220, −00123).
   4. A 5 digit non-negative integer M in decimal format without sign, such that N+M represents the upper bound of the safe measurement range (e.g. 12345, 00020).

   For the purposes of this question, additionally assume that the system will reject as invalid any integer input for N or M that is less than −32768 or greater than 32767 (i.e. outside the representable range of a signed 16-bit integer).

   You have been tasked with writing tests for a deployed DWS system with one registered sensor: "smoke concentration", and one registered alert: "fire".

   (a) For the DWS described above, determine the equivalence classes for the above FOUR inputs.

   (5 marks)

   (b) Determine the boundaries of the equivalence classes identified in your answers to Q4(a). For each boundary, identify a value on the boundary, a value just below the boundary, and a value just above the boundary.

   (4 marks)

   (c) You intend to perform **defensive testing** of the link function for the DWS described above. Design a set of test cases to test the FOUR inputs based on equivalence classes and boundary values identified in your answers to Q4(a) and Q4(b).

   (8 marks)

   (d) Imagine that the system is changed to additionally reject any input such that N+M is outside the representable range of a signed 16-bit integer. Briefly discuss how this would complicate the definition of the equivalence classes of M and the test suite of Q4(c).

   (4 marks)

   (e) Imagine that you have been asked by your manager to design a single generic set of test cases for the link function that is applicable to every possible deployed DWS system.

      i. Explain why it would be challenging to perform defensive testing based on equivalence classes using such a test suite.

      (2 marks)

      ii. Propose a potential solution for this challenge.

      (2 marks)



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## Appendix A

The Disaster Warning System (DWS)

The Disaster Warning System (DWS) is a platform for collecting, managing, and reacting to data related to natural disasters, with the aim of predicting and preventing them. It is intended to be deployed by a local government and connected to multiple sensors and alert systems that may be present in the vicinity. Since each local government may have different sensor data sources available, and different alert systems relating to the tracking of particular natural disasters, the DWS must flexibly adapt to whatever is provided.

The DWS is configured with a list of registered sensors, and a list of registered alert systems. An appointed DWS manager can add and remove sensors and alert systems to/from the registration lists in the DWS software - the DWS will accept this new registration so long as it can detect that the registered system is physically connected to the DWS. Once a sensor has been registered, it can send measurement updates to the DWS at any time, at its own discretion. All measurement updates consist of a single integer value representing the measurement taken by the sensor. The manager can further configure the DWS to link some registered sensor to some registered alert, so that the alert is triggered when the sensor reports a value outside a certain range - or unlink a link that already exists. For example, the city of San Angeles may purchase and install a DWS which they connect to a forest smoke sensor and a fire brigade alert, and configure the DWS to send a "forest fire" alert to the fire brigade if the sensor sends a reported measurement of smoke particles occurring in its nearby forest at a concentration of more than 1000 parts-per-million.

The DWS should maintain a full record of system interactions which update the registration and/or linking of a sensor or alert system, along with a record of all messages the DWS has received from sensors and sent to alert systems. For each registered sensor, if the DWS has not received an update from that sensor in over 24 hours, an alert should be sent to the manager. The manager can also request the full record of system interactions and messages at any time.

END OF PAPER



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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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